

Annex B

St. Andrew's Junior College

Qn/No	Topic Set	Paper Number	Answers	Marks
	Pure Maths			
1	Logarithm	8863	$x = \frac{7}{2}$ or $x = -1$ (reject)	4
2	Functions	8863	a) $R_f = D_g \therefore gf$ exist $gf : x \mapsto 1 + \ln(x+1), x \in \mathbb{R}, x > -1$ b) $f^{-1} : x \mapsto e^{-x} - 1, x \in \mathbb{R}$	7
3	Graphing Techniques & Inequalities	8863	asymptotes : $x = 2, y = -4$ axis intercepts : $\left(\frac{3}{4}, 0\right) \left(0, -\frac{3}{2}\right)$ - B1 $(-\sqrt{2}, 0), (\sqrt{2}, 0), (2, 0)$ - B1 Ans: $x < 1$ or $x > 2$	6
4	Application of Differentiation	8863	a ii) $y = \frac{1}{2}x + 3$ bi) $\frac{40}{73}$ cm / s or 0.548 cm / s ii) $x = -7.97$ (reject) or $x = 5.30$ (3s.f) iii) max. $V = 445 \text{ cm}^3$	13
5	Integration and it's application	8863	a) $5 \ln x+2 + C; \frac{2x^3}{3} - \frac{1}{6}x^2 + C$	10
	Statistics			
6	Sampling	8863		6
7	Normal Distribution	8863	i) 0.841 ii) 0.529 iii) 4.255 iv) 0.7584	10
8	Binomial and Sampling Distribution	8863	i) 0.382 ii) 0.245 ; mean = 2, s.d = 1.095	10
9	Probability	8863	ii) 0.5 iii) 0.554 iv) 0.69	11
10	Hypothesis Testing	8863	a) $p = 2.04 \times 10^{-5}$ b) 48.01 $p = 0.00571$	13
11	Linear Regression	8863	i) $r = 0.951$ ii) $y = 0.11469x + 16.00824$ iii) 80 mins iv) Estimated cost \$7.11	10

SAINT ANDREW'S JUNIOR COLLEGE

Preliminary Exam

MATHEMATICS Higher 1

8863

Wednesday

27 August 2008

3 hours

Additional materials : Answer paper
Cover Sheet
MF15

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions. Total marks : 100

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically state otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematic steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of 5 printed pages including this page.

[Turn over

Section A: Pure Mathematics [40 marks]

- 1 Solve for x using an algebraic method, given that [4]
 $\log_2(2x+1) + \log_2(x-3) = \log_4 16$
- 2 The functions f and g are defined by
 $f : x \mapsto \ln(x+1), \quad x \in \mathbb{R}, x > -1$
 $g : x \mapsto 1+x, \quad x \in \mathbb{R}$
- a) Explain why gf exist and define gf in a similar form. State also it's range. [5]
- b) Define f^{-1} in a similar form. [2]
- 3 Sketch the graphs of $y = \frac{4x-3}{2-x}$ and $y = |x^2 - 2|$ on a single diagram
 State the coordinates of the points of intersection with the axes and the equations of the asymptotes.
 Hence solve the inequality $\frac{4x-3}{2-x} < |x^2 - 2|$ [6]
- 4 (a) A curve is represented by the equation $y = \left(1 + \frac{2}{x}\right)^2$, where $x > 0$
 The point P lies on the curve where $x = 2$.
- i) Find $\frac{dy}{dx}$ and hence show that the gradient of the curve at P is -2 [3]
- ii) Find the equation of the normal to the curve at P . [3]
- 4 (b) Water is poured into a fish tank at a rate of $20 \text{ cm}^3/\text{s}$. The volume, $V \text{ cm}^3$, of the water in the fish tank, when the height of the water is $x \text{ cm}$, is given by,
- $$V = 0.3x^3 + 1.2x^2 + 2x + 5$$
- Find
- i) the rate of increase in the height of the water when $x = 5$ [3]
- ii) the height of the water when the rate of increase in the height is 0.5 cm/s [2]
- iii) Show V is an increasing function and hence find the maximum value that V can take, given that $6 \leq x \leq 10$ [2]

5 (a) Find $\int \frac{5}{x+2} dx$ and $\int \left(2x^2 - \frac{1}{3}x \right) dx$ [2]

- (b) Sketch the curves of $y = \frac{5}{x+2}$ and $y = 2x^2 - \frac{1}{3}x$. On the same diagram, stating clearly the asymptotes, if any, find the coordinates of the point of intersection of the 2 curves.
Hence, show that the area of the region enclosed by the 2 curves and the y axis is $5\ln 3 - 5\ln 2 - \frac{1}{2}$. [8]

Section B: Statistics [60 marks]

- 6 A company, with 9320 employees, provides refuse collection services for 47 councils in a particular county. The company asks a market research firm to carry out an opinion poll of its employees concerning union membership.

- (i) Describe how the market research firm could obtain a simple random sample of size 120 from the 9320 employees. [3]
- (ii) It is proposed that a stratified sample be used.
- (a) Suggest two factors which could be used to stratify the sample. [2]
- (b) Give a reason why the market research firm might prefer a stratified sample to a random sample. [1]

- 7 When Monica walks to work from home, she uses either route A or route B. Her journey time, X minutes, by route A may be assumed to be normally distributed with a mean of 37 and a standard deviation of 8. Determine:

- (i) $P(X < 45)$, [1]
- (ii) the probability that the total time taken for 3 journeys by route A is more than 110 minutes. [3]

Her journey time, Y minutes, by route B may be assumed to be normally distributed with a mean of 40 and a standard deviation of σ .

- (iii) Given that $P(Y > 45) = 0.12$, show that $\sigma = 4.255$. [3]
- (iv) Monica made 10 journeys using route B. Find the probability that the mean journey time is between 39 and 43 minutes. [3]

- 8 Each weekday, Monday to Friday, Trina catches a train from her local station. She claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day. Assuming her claims to be true, determine the probability that the train arrives on time at the station:
- (i) on at most 3 days during a 2-week period (10 days); [2]
 - (ii) on more than 10 days but fewer than 20 days during an 11-week period (55 days) using normal approximation. [3]
- Assuming Trina's claims to be true, determine the mean and standard deviation for the number of times during a week (5 days) that the train arrives on time at the station. [2]
- Each week, for a period of 13 weeks, Trina's travelling colleague, Suzie, records the number of times that the train arrives on time at the station. Suzie's results are 2, 2, 4, 1, 2, 3, 3, 2, 2, 0, 3, 2, 0. [3]
- Calculate the mean and standard deviation of these values.
Hence comment on the likely validity of Trina's claims.
- 9 A hotel offers guests a choice of a full English breakfast or a continental breakfast. The probabilities of these choices being made are 0.4 and 0.6 respectively. The probability that a guest will choose the same type of breakfast as the day before is half the probability of that type of breakfast on previous day. A guest, Alan, had breakfast at the hotel for 3 consecutive days.
- (i) Draw a tree diagram to illustrate all his possible choices of breakfast on these 3 days. [2]
 - (ii) Find the probability that he had English breakfast on the second day [2]
 - (iii) Find the probability that he had continental breakfast on 2 of the 3 days. [3]
 - (iv) Alan had English breakfast on the second day. Find the probability that he had continental breakfast on the third day. [4]

- 10 A stretch of dual carriageway in a school zone has a speed limit of 30 km/h. Before the introduction of speed cameras, the speeds of vehicles using this dual carriageway had a mean of 41 km/h and a standard deviation of 8.5 km/h. Following the introduction of speed cameras on this dual carriageway, ten drivers were prosecuted for exceeding the speed limit. Their recorded speeds, in km/h, were: 53.7, 39.9, 46.0, 62.8, 44.9, 55.3, 49.6, 48.2, 53.0 and 66.9.
- (a) Lim, a representative of a motoring organisation, stated that he had examined these data and found significant evidence that the mean speed of vehicles had increased since the introduction of the speed cameras. He therefore claimed that the road would be much safer if the cameras were removed.
- (i) Show that, if the ten recorded speeds are regarded as a random sample from a normal distribution with standard deviation 8.5, there is evidence, significant at the 1% level, that the mean of this distribution exceeds 41. [5]
- (ii) Explain why Lim's claim is not valid. [2]
- (b) To investigate whether the mean speed of vehicles has increased since the introduction of speed cameras, the speeds of a random sample of 120 vehicles using the dual carriageway are recorded and are found to have a mean of 31.6 and a standard deviation of 6.9. [6]
- Use this second sample, and the 5% significance level, to examine whether there is evidence that the mean speed of vehicles now exceeds 30 km/h.
- 11 Stan is a retired teacher who supplements his pension by mowing lawns for customers who live nearby. As part of a review of his charges for this work, he measures the areas, $x \text{ m}^2$, of a random sample of eight of his customers' lawns and notes the times, y minutes, that it takes him to mow these lawns. His results are shown in the table.
- | | | | | | | | | |
|-----------------------|----------|----------|----------|----------|----------|----------|----------|----------|
| Customer | <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> | <i>E</i> | <i>F</i> | <i>G</i> | <i>H</i> |
| Area, $x \text{ m}^2$ | 360 | 140 | 860 | 600 | 1180 | 540 | 260 | 480 |
| Time, y mins | 50 | 25 | 135 | 70 | 140 | 90 | 55 | 70 |
- (i) Plot a scatter diagram of these data and comment on the relationship between x and y . [2]
- (ii) Calculate the equation of the least squares regression line of y on x . [2]
- (iii) Estimate the time needed, to the nearest integer, for mowing a customer's lawn that has an area of 560 m^2 and comment on the reliability of your estimate. [2]
- (iv) If Stan decides to follow the model in (ii) and given that Stan charges \$6 per hour, estimate how much more or less customer *H* has to pay. [4]

- End of paper -